

VANTA-1T

An Analytical Single-Package Inference Architecture for Trillion-Parameter Mixture-of-Experts Models

68.1%

less resident weight memory
vs modeled MXFP4

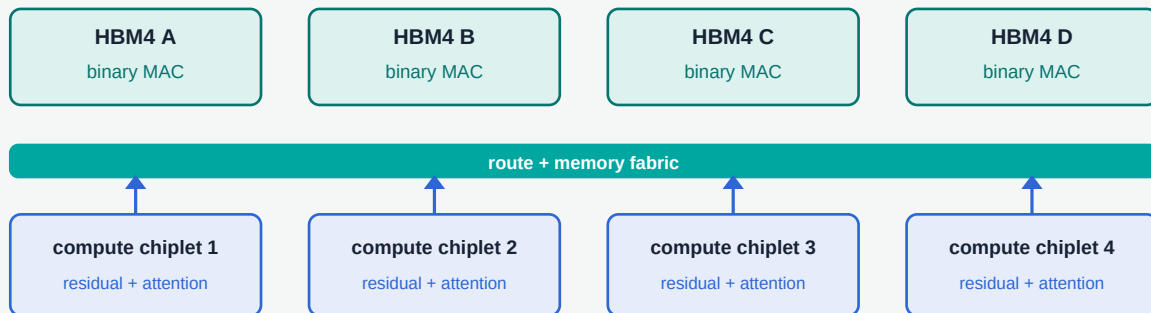
180.1 GB

stretch weight image
inside 192 GB target

1T / 32B

total / active parameters
reference workload

VANTA-1T package dataflow



Top-k is resolved before expert movement; higher precision work stays on compute chiplets.

STATUS: capacity-plausible, quality-unproven. No fabricated silicon and no measured performance.

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Abstract

This paper presents VANTA-1T, an inference-only accelerator concept for a one-trillion-parameter sparse mixture-of-experts (MoE) language model with approximately 32 billion parameters active per token. The objective is not to claim fabricated hardware, but to test whether a trillion-parameter model can be made resident in a single accelerator package while reducing its resident weight image by more than 50 percent relative to a transparent MXFP4-style baseline. VANTA-1T represents routed-expert matrices as a one-bit sign backbone, group scales, and a structured 4-bit low-rank residual. Higher-risk operations - attention, routing, embeddings, the shared expert, and the dense layer - remain at higher precision. A custom HBM4 base die executes binary dot-products near the weights; four logic chiplets apply residual corrections, attention, routing, and scheduling. For the public Kimi K2.5 configuration, the resulting analytical resident weight image is 180.1 GB, compared with 563.7 GB for the modeled MXFP4 baseline, a 68.1 percent reduction. Four 48 GB HBM4 stacks provide 192 GB, leaving 11.7 GB after an 8K-context single-user 8-bit MLA KV cache. A first-order bandwidth model gives a 701.5 token/s batch-1 decode ceiling at 8K context under an explicitly assumed 35 percent service-efficiency factor. These are calculated capacities and ceilings, not silicon measurements. The central gating experiment is whether the binary-residual representation preserves model quality. Physical area, timing, thermals, yield, cost, and end-to-end throughput remain unvalidated.

1. Scope and research question

Large sparse MoE models create a useful asymmetry: only a small fraction of the weights participate in each token, yet the full expert population normally remains resident somewhere in the serving system. The public Kimi K2.5 model is a concrete example: Moonshot AI reports one trillion total parameters and 32 billion active parameters, with 384 routed experts and eight selected experts per token. This makes capacity a system-level problem even when per-token arithmetic is much smaller than the total model.

VANTA-1T asks a narrow engineering question:

Under an aggressive but explicit weight representation, can the complete expert population of a 1T/32B-active MoE fit in one accelerator package, and can the resident weight image be at least 50 percent smaller than an MXFP4-style representation?

The paper answers only the analytical capacity question. It does not claim a new trained model, tapeout-ready RTL, a manufacturing process, or measured performance. The intended contribution is an auditable architecture target, an executable first-order model, and a falsifiable validation plan.

2. Reference workload and public hardware facts

The workload parameters are taken from the public Kimi K2.5 configuration: 61 transformer layers, one dense layer, 384 routed experts, eight selected experts, hidden width 7168, routed-expert intermediate width 2048, MLA latent rank 512, and maximum context 262,144 tokens. Treating each routed SwiGLU expert as gate, up, and down matrices gives:

$$P_{\text{routed}} = 60 \times 384 \times 3 \times 7168 \times 2048 = 1.014686 \times 10^{12} \text{ weights.}$$

This derived routed-expert count is close to the published one-trillion total parameter description. The small mismatch is expected because the public total is rounded and the architecture contains non-expert parameters.

For the memory envelope, the research target uses four 48 GB HBM4 stacks and 2.8 TB/s per stack. These values follow Micron's public 48 GB 16-high sample and greater-than-2.8-TB/s product statements; the analysis deliberately avoids using a faster peak claim. For context, NVIDIA's preliminary Rubin data lists 288 GB HBM4 and 22 TB/s per Rubin GPU, and four Rubin GPUs in the NVL4 module. Those public figures are references, not measured VANTA baselines.

OpenAI's Jalapeno report motivates three system principles used here: keep model state and KV state close to compute, treat the network as part of the accelerator, and activate resources differently for compute-heavy prefill and bandwidth-heavy decode. VANTA-1T independently instantiates those principles around an explicit binary-residual expert representation.

3. Architecture

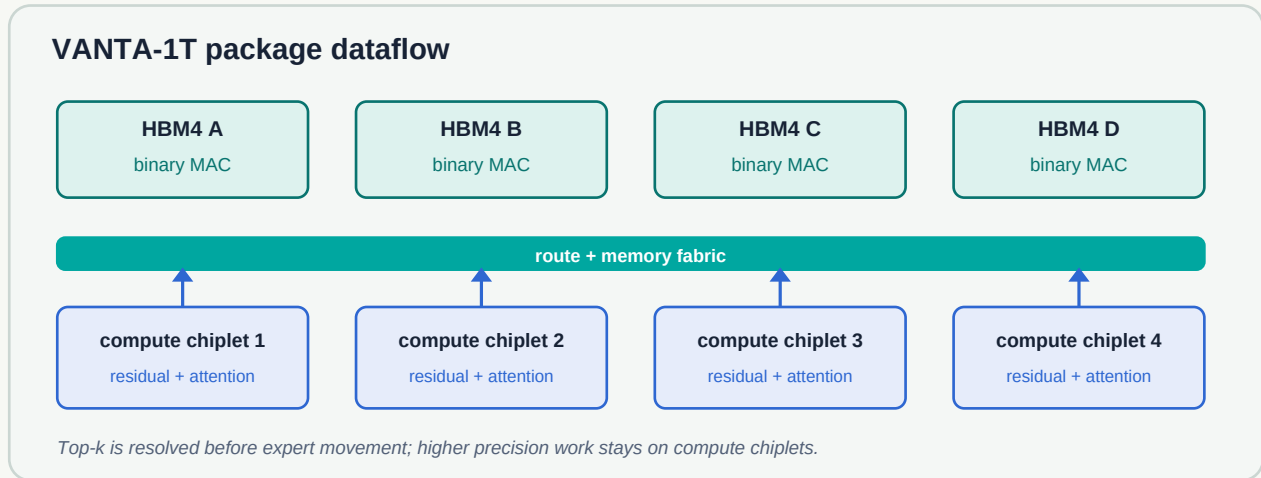


Figure 1. Package-level split between near-memory binary expert work and higher-precision chiplet work.

3.1 Package organization

The research configuration contains one accelerator package, four compute chiplets, four custom 48 GB HBM4 stacks, a silicon interconnect fabric, and a target of 512 MB distributed SRAM. The raw capacity is 192 GB and the raw HBM bandwidth is at least 11.2 TB/s under the conservative per-stack assumption. The power target is 650 W, but no thermal proof is claimed.

A production-capacity option expands to six 48 GB stacks, 288 GB capacity, 16.8 TB/s raw bandwidth, and a 780 W package target. This option is intended for practical concurrency and runtime reserve, not to improve the headline weight-compression ratio.

3.2 Split execution path

Each routed expert is decomposed into two computational paths:

1. Binary backbone. The sign bit of each expert weight remains adjacent to the weight's HBM channel. XNOR/popcount-like dot-product lanes on a custom base die form a coarse expert output.
2. Residual correction. Structured low-rank residual factors, stored at 4-bit precision, are prefetched into chiplet SRAM and applied on conventional vector/matrix tiles.

The attention blocks, router, embeddings, shared expert, and dense layer stay at higher precision inside a separate 12 GB resident allowance. This prevents the most routing- and activation-sensitive operations from being forced into the binary path.

The base-die computation is a research proposal. It may require a custom HBM logic die or a tightly coupled memory-side companion die and should not be interpreted as a feature of commodity HBM4.

3.3 Route before movement

The router produces top-k expert identifiers before expert data movement. A compiler places experts from an observed traffic matrix rather than from numeric expert IDs. The objective is to minimize cross-domain traffic while avoiding HBM-channel hot spots. Recently used residual factors occupy a small victim cache. A worst-case network path remains mandatory because average locality can fail under adversarial prompts or distribution shift.

3.4 Prefill and decode

During prefill, token blocks are spread over the four compute chiplets. Attention and router logits execute on matrix/vector tiles; expert IDs are multicast to the appropriate memory sectors; binary backbones run beside the resident weights; and residual corrections overlap with the next expert prefetch.

During decode, a latency-oriented scheduler pins a request, its hot MLA KV blocks, and its likely experts to one locality domain. Binary backbone, residual correction, and attention are pipelined. The same physical tiles change scheduling mode between prefill and decode instead of requiring two separate accelerator classes.

4. Weight representation

The stretch profile assigns the routed experts:

Term	Effective bits per routed weight
One-bit sign backbone	1.000
One 8-bit scale per 64 values	0.125
4-bit residual over 4 percent equivalent parameters	0.160
Packing and control metadata allowance	0.040
Total	1.325

The 4 percent residual budget is an architecture allocation, not evidence that 4 percent is sufficient. It can be implemented by low-rank factors, sparse outliers, or a hybrid chosen per expert. An expert-sensitive compiler may assign different residual ranks while preserving the global budget.

The MXFP4-style comparison uses 4.0 payload bits, one 8-bit scale per 32 weights (0.25 bit/weight), and a 0.10 bit/weight metadata allowance, for 4.35 effective bits. Both representations add the same 12 GB non-expert allowance. This makes the comparison reproducible, but it is not a claim about the exact in-memory layout of every deployed MXFP4 implementation.

5. Analytical model

5.1 Resident weights

For routed parameter count P , effective expert precision b , and non-expert allowance M_{other} :

$$M_{\text{weights}} = P \cdot b / 8 + M_{\text{other}}.$$

Using $P = 1.014686 \times 10^{12}$ and $M_{\text{other}} = 12 \text{ GB}$ gives 563.736 GB for 4.35 bits and 180.057 GB for 1.325 bits.

5.2 KV cache

The simplified 8-bit MLA cache model stores the latent vector and RoPE component per layer, token, and sequence:

$$M_{\text{KV}} = L \cdot T \cdot S \cdot (r_{\text{KV}} + d_{\text{RoPE}}) \cdot b_{\text{KV}} / 8.$$

For 61 layers, rank 512, RoPE dimension 64, 128K tokens, and one sequence, the cache is 4.605 GB. The model excludes allocator fragmentation, temporary activations, prefix metadata, and runtime workspaces.

5.3 Decode ceiling

The single-stream bandwidth ceiling is:

$$\text{tokens/s} \leq \eta \cdot B_{\text{raw}} / (M_{\text{active_weights_per_token}} + M_{\text{KV_read_per_token}}),$$

where η is the end-to-end service-efficiency assumption. The default η is 0.35. It absorbs protocol overhead, imbalance, bubbles, non-HBM operations, and imperfect overlap. It is a scenario parameter, not a measurement.

6. Results

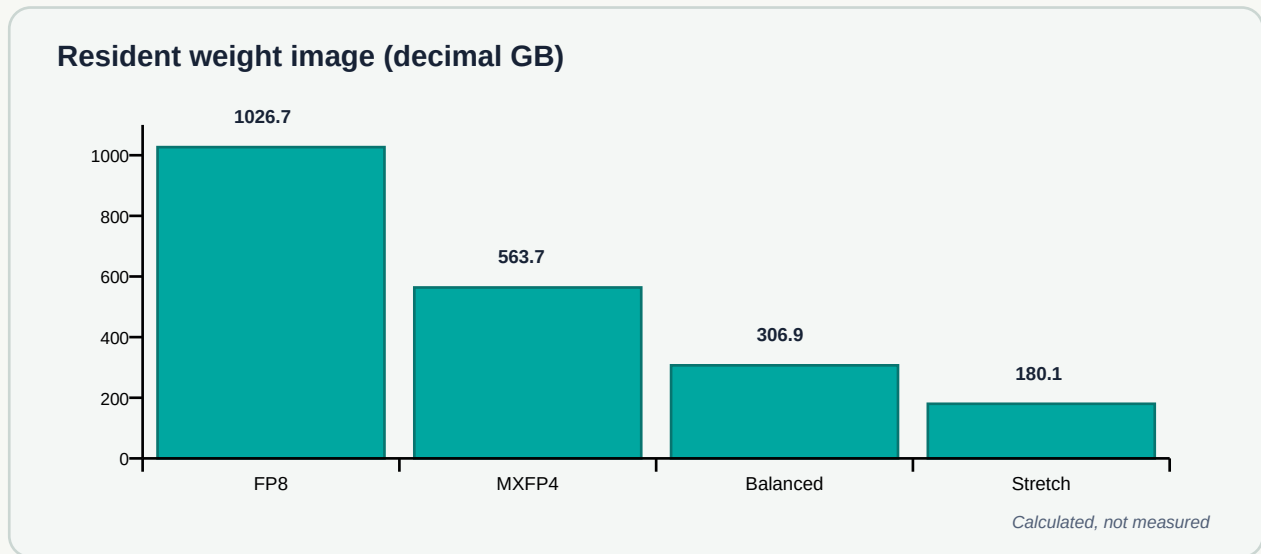


Figure 2. Analytical resident weight image. VANTA profiles are quality hypotheses; the values are not silicon measurements.

6.1 Memory footprint

Representation	Effective expert bits	Resident weights	Change vs MXFP4
FP8 reference	8.00	1,026.7 GB	+82.1 percent
MXFP4-style baseline	4.35	563.7 GB	baseline
VANTA balanced	2.325	306.9 GB	-45.6 percent
VANTA stretch	1.325	180.1 GB	-68.1 percent

The balanced profile does not meet the 50 percent target. The binary-residual stretch profile exceeds it by 18.1 percentage points. The conclusion is capacity-plausible only if its quality hypothesis survives evaluation.

6.2 Representative serving cases

Configuration	Weight GB	KV GB	Total GB	Capacity GB	Headroom GB	Single-stream ceiling (a)
4 stacks, 8K, 1 user	180.057	0.288	180.345	192	11.655	701.5 tok/s
4 stacks, 128K, 1 user	180.057	4.605	184.663	192	7.337	395.7 tok/s
6 stacks, 128K, 8 users	180.057	36.843	216.900	288	71.100	593.6 tok/s

The four-stack target demonstrates residency but leaves little operational margin. The six-stack option is the more credible serving system. The decode figures are bandwidth ceilings under the 35 percent assumption and should not be presented as benchmark results.

(a) The eight-user row still reports the one-stream equation at that context and bandwidth. The first-order model does not predict per-user or aggregate throughput under batching, expert reuse, or bandwidth sharing.

6.3 Footprint claim boundary

The result supports one strong, narrow claim: the analytical resident weight image is 68.1 percent smaller than the stated MXFP4-style baseline. It does not show 68.1 percent smaller silicon or rack area.

As a secondary count comparison, VANTA uses one accelerator package while NVIDIA describes four Rubin GPUs in NVL4, a 75 percent reduction in accelerator package count. This is not an area, power, cost, or performance comparison. One Rubin GPU also has enough published capacity to hold the VANTA weight image, so the package-count comparison should never replace the primary memory representation result.

7. Validation plan

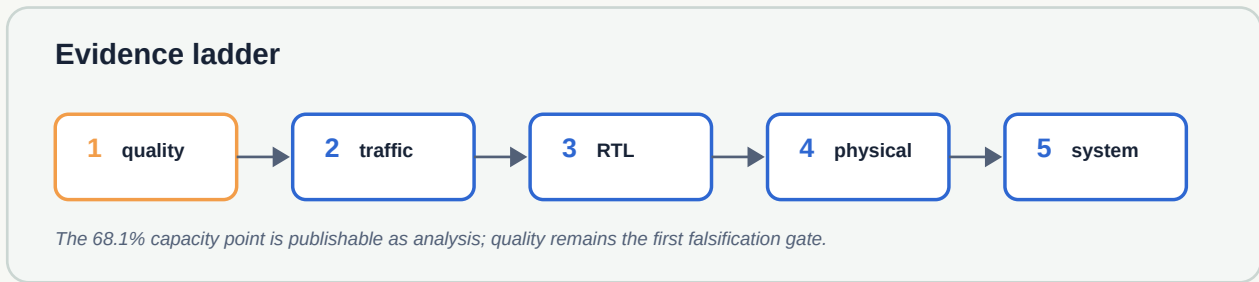


Figure 3. Validation sequence from the first quality falsification gate to end-to-end system evidence.

The following gates turn the proposal into an engineering result:

1. **Quality gate.** Quantize a smaller open MoE proxy with the same binary plus residual method. Compare perplexity, task accuracy, reasoning, routing stability, and vision quality against BF16, INT4, and MXFP4.
2. **Scaling gate.** Fit residual-rank allocation and distillation on a larger MoE. Report the Pareto curve of quality versus effective bits, including failed runs.
3. **Traffic gate.** Implement a cycle-accurate simulator with token traces, expert imbalance, HBM channels, SRAM banks, NoC contention, and prefill/ decode transitions.
4. **RTL gate.** Compile and formally check the router and binary-residual datapath. The included RTL is pedagogical and has not yet passed a production HDL toolchain.
5. **Physical gate.** Use a real PDK for synthesis, place-and-route, package escape, signal integrity, power delivery, thermals, yield, and cost.
6. **System gate.** Measure end-to-end request traces with batching, speculative decoding, prefix reuse, failures, and service-level objectives.

Failure of Gate 1 falsifies the 1.325-bit profile, but not the framework: the executable model exposes precision as a parameter so a higher-quality point can be substituted. The balanced 2.325-bit point, for example, reduces memory by 45.6 percent and therefore misses the project's stated stretch target.

8. Related work and position

Recent work makes clear that the components of VANTA-1T have active research precedent. Expert-wise mixed precision uses router-derived sensitivity to allocate quantization precision. MoBiE studies binarization of MoE models and uses low-rank structure and routing-stability objectives. Expert Streaming examines on-chip expert streaming across chiplets, while Sieve studies dynamic GPU and HBM-PIM scheduling. Work on phase-disaggregated inference argues that prefill and decode benefit from different precision and datapath choices, and ELDR reports structured expert reuse.

VANTA-1T is therefore positioned as an open architectural synthesis and a specific trillion-parameter capacity target, not as a claim that binarization, low-rank residuals, near-memory compute, or phase-aware scheduling were individually invented here. Its useful novelty, if validated, would be the co-designed combination and its demonstrated quality-capacity-performance operating point.

9. Threats to validity

- Quantization quality is entirely unmeasured on Kimi K2.5.
- A one-trillion-weight training or distillation run may be economically inaccessible to an independent project.
- The non-expert 12 GB allowance is a coarse estimate and may be too small.
- Four HBM stacks leave insufficient reserve for a robust runtime.
- The 35 percent service efficiency can overstate or understate real systems.
- Expert imbalance and network contention are not represented in the simple bandwidth equation.
- Custom logic under HBM changes thermals, yield, verification, and supplier constraints.
- No PDK-backed physical area exists, so the user's original physical-space ambition remains unproven.

10. Claim ledger

Claim	Status	Required evidence
68.1 percent smaller resident weight image than modeled MXFP4	Calculated	Public config plus executable equations
Fits 4 x 48 GB HBM at 8K, one user	Calculated	Capacity model; runtime reserve omitted
701.5 token/s at 8K	Modeled ceiling	Cycle simulator and silicon benchmark
Preserves useful model quality	Unvalidated	QAT/distillation and evaluation suite
650 W package	Target	Power and thermal implementation
More than 50 percent smaller physical package	Unknown	PDK floorplan and package study
Cheaper than current accelerators	Unknown	Yield, packaging, memory, and system BOM

11. Conclusion

VANTA-1T provides a falsifiable path to resident single-package inference for a 1T/32B-active MoE. Its stretch representation reduces the calculated resident weight image from 563.7 GB to 180.1 GB, exceeding the 50 percent memory target with a 68.1 percent reduction. Four 48 GB stacks are sufficient only as a tight research configuration; six stacks are the practical capacity option. The design's most important next step is not a louder hardware claim but a quality experiment. If a binary backbone plus a 4 percent 4-bit residual budget cannot preserve useful model behavior, the headline point fails. If it does, the combination of route-before-movement, near-memory binary compute, and phase-aware chiplet scheduling becomes a serious candidate for deeper cycle-level and physical design.

References

1. Moonshot AI. Kimi K2.5 model card and configuration. 2026. <https://github.com/MoonshotAI/Kimi-K2.5>
2. OpenAI. Jalapeno: First results. 2026. <https://openai.com/index/jalapeno-first-results/>
3. NVIDIA. Vera Rubin NVL72 platform specifications. 2026. <https://www.nvidia.com/en-us/data-center/vera-rubin-nvl72/>
4. Micron. HBM4 product page. 2026. <https://www.micron.com/products/memory/hbm/hbm4>
5. Samsung Semiconductor. HBM4 product page. 2026. <https://semiconductor.samsung.com/dram/hbm/hbm4/>
6. Efficient Quantization of Mixture-of-Experts Models. arXiv:2604.06515, 2026. <https://arxiv.org/abs/2604.06515>
7. Expert Streaming: Multi-Chiplet MoE Inference. arXiv:2603.27624, 2026. <https://arxiv.org/abs/2603.27624>
8. Sieve: Dynamic GPU and HBM-PIM Scheduling for MoE Inference. arXiv:2605.11277, 2026. <https://arxiv.org/abs/2605.11277>
9. MoBiE: Mixture-of-Experts Binarization. arXiv:2604.06798, 2026. <https://arxiv.org/abs/2604.06798>
10. HBM Is Not All You Need: Efficient Disaggregated LLM Serving across Memory-heterogeneous Accelerators. arXiv:2606.29986, 2026. <https://arxiv.org/abs/2606.29986>
11. ELDR: Expert-Locality-Aware Decode Routing for PD-Disaggregated MoE Serving. arXiv:2607.00466, 2026. <https://arxiv.org/abs/2607.00466>

Authorship and AI assistance

Mahee Monjur is the author and accepts responsibility for all claims. OpenAI models assisted with source discovery, architecture exploration, analytical modeling, implementation, checking, visualization, and drafting. AI systems are not authors.

Artifact availability

The accompanying artifact contains the executable Python model, generated JSON and CSV outputs, tests, reproducible figures, pedagogical SystemVerilog blocks, the interactive HTML architecture lab, and the publication materials. Code is released under the MIT license; the paper and figures are released under CC BY 4.0.